Australia has the World's Best Petrol Price Data: FuelWatch and FuelCheck

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Abstract

This article introduces two retail fuel price databases in Australia, FuelWatch and FuelCheck. Each dataset consists of a complete panel of station-level prices over time and across stations, and they are available for public access via web download. We provide background on how the two databases were established, describe their data generating processes, overview some existing studies that utilise these data, and provide suggestions for future potential use.

1. Introduction

FuelWatch and FuelCheck are legislated retail fuel price platforms operated by the state governments of Western Australia and New South Wales, respectively. Their websites[1] display real-time information on petrol prices across stations to help consumers find the lowest-price stations. Luckily for researchers and students in economics, especially those studying industrial organisation, these platforms also make these real-time station-level data publicly available, and they can be readily accessed via web download. Importantly, both datasets are complete: they contain every price charged at every point in time at every station since the respective launches of FuelWatch (Jan 2001) and FuelCheck (Sept 2016) through to present day.

At least in the context of industrial organisation, there are many core issues that can be examined with these unique datasets, including collusion, price discrimination, mergers, consumer search, vertical relations, and the dynamics of cost pass-through (Eckert 2013). Because of the breadth of topics that can be examined with petrol price data, this makes retail petrol markets an ideal laboratory for empirically examining problems in industrial organisation.

In this paper, we provide a detailed overview of the data generating processes and contents of FuelWatch and FuelCheck databases. We describe how the data are collected and published on each platform, followed by a comprehensive description of the variables contained in each dataset. In addition, we present some basic patterns and trends in the data at different levels of aggregation and time frequency, from monthly market level, to station-minute level. Along the way, we overview research in economics which have exploited these rich data for empirically examining issues of market power, and point to future areas of research that could be explored.

2. Background on FuelWatch and FuelCheck

The Australian petrol industry operates at three levels: refining and importing, wholesale, and retail. About 60% of unleaded petrol sold in Australia is refined by local refineries, and the remainder is imported from

[1] (FuelWatch) https://www.fuelwatch.wa.gov.au/ ; (FuelCheck) https://www.fuelcheck.nsw.gov.au/

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overseas. As a significant importer of petrol from the Asia Pacific region, the level of retail prices day-to-day are largely determined by Singapore's Mogas 95 Unleaded Petrol (MOPS95) price, which is determined in the global crude oil market.

The retail sector for fuel in Australia is highly concentrated and is dominated by few major oil companies (BP and Caltex) and the two major supermarket brands (Coles and Woolworths). In total, dominant players in this industry serve approximately 80 percent of the market, with mid-tier and independent retailers serving the remaining 20 percent (ACCC, 2007, 2017).

Beyond their concentration, petrol retailers are able to exercise market power because consumers face "search costs," which are any costs associated with finding the lowest-price station at a given time (Eckert 2013). These costs may involve, among other things, the fuel cost from traveling from station-to-station to shop for the lowest petrol price, or the opportunity cost of time from shopping for petrol. The larger the search costs consumers face, the greater market power petrol stations enjoy.

Historically, rising fuel prices and concerns of market power have led to several public investigations into the nature of competition in major cities and regional locations. In WA and NSW, state governments have taken an even more proactive approach, and created the FuelWatch and FuelCheck price transparency platforms to reduce consumer search costs and give motorists the power to make informed petrol purchase decisions.

2.1 FuelWatch

FuelWatch (https://www.fuelwatch.wa.gov.au/) is the government-operated price transparency platform for motorists in Australia. Its main purpose is to monitor and report on Western Australia's wholesale and retail fuel prices under the Petroleum Products Pricing Act 1983. The platform was launched in January 2001 by the WA government in response to consumers' concerns about intra-day price fluctuations and the significant difference between city and regional fuel prices (FuelWatch, 2011). It is a unique service that enables motorists in Western Australia to identify fuel prices at all stations on their personal computer or on a mobile device. Users can further specify specific postcodes with which to list prices to refine their online price search behaviour. Figure 1 depicts output from the FuelWatch platform for Perth's CBD, postcode 6000.

When FuelWatch was initially launched, its geographical boundaries included towns with population of more than 5000 people and at least two fuel retail outlets. This of course included the entire metropolitan area of Perth, as well as many regional areas from Albany to Geraldton. Today, FuelWatch extends beyond these areas and now includes approximately 80% of regional outlets in WA (FuelWatch, 2011).

How has FuelWatch been able to provide real-time information as early as 2001, in the early days of the Internet? It does so through a two-part daily price setting rules established by the state government as part of the FuelWatch price transparency program:

1. Each day at 2pm, retailers submit to the government via web upload all of the petrol prices that will be charged at their stations tomorrow.

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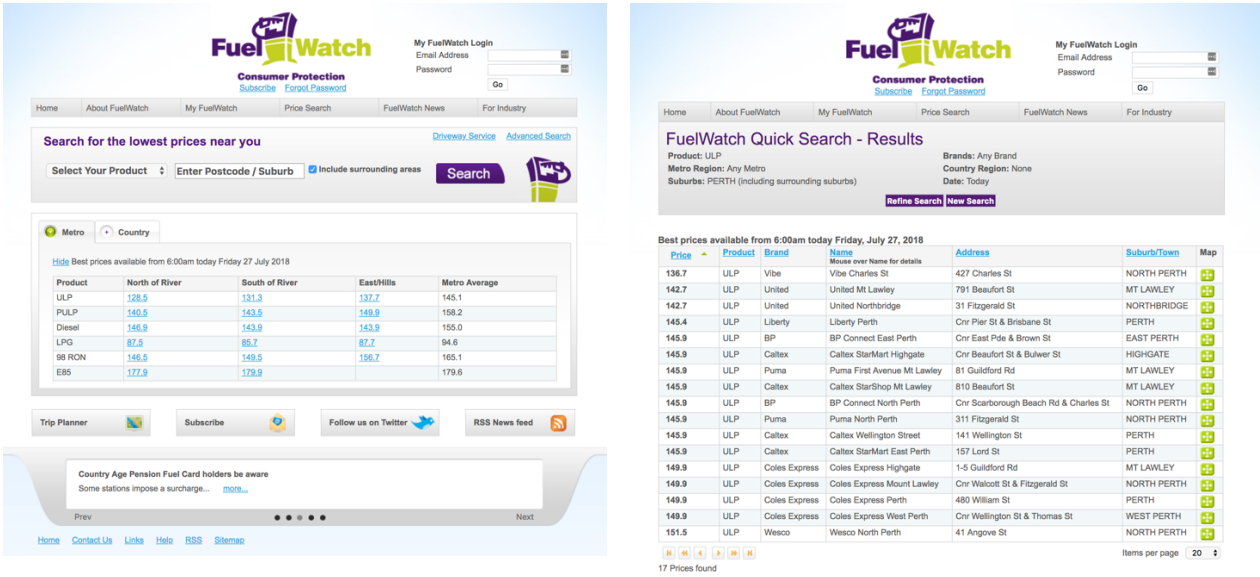
2. When the stations open at 6am the next day, they are required by law to post the prices that they uploaded at 2pm the day before, and they must keep prices at these levels for 24-hours. In practice, this is known as the **24-hour rule**.

An implication of these price setting rules is that stations effectively set prices simultaneously once each day. Using these data, FuelWatch is able to provide online the entire distribution of prices currently in the market before 2pm each day. After 2pm each day, FuelWatch is able to provide both the today's entire distribution of prices as well as tomorrow's distribution of prices. This information thus helps consumers find the cheapest stations on a given day, and potentially helps them to avoid paying higher prices tomorrow, should prices be on the rise.

Figure 1: FuelWatch Home Page and Search Results for Perth ULP Prices (Postcode 6000)



2.2 FuelCheck

FuelCheck (https://www.fuelcheck.nsw.gov.au/) was more recently introduced in August 2016 by the NSW government. It has full geographic coverage of the state and posts data online for all types of fuel offered across stations at a point in time. Like FuelWatch in WA, FuelCheck also allows users to search for petrol prices within particular postcodes.

While the high-level goals of FuelCheck and FuelWatch are similar, their respective implementations in terms of data collection and online price post differ. In particular, the FuelCheck program does not put any restrictions on the timing of price changes and reporting by retailers. Rather, retailers make a station-level price change through the FuelCheck website whereby they identify the relevant station they are changing their price at, change the fuel price, and confirm their update online. Price changes at the station are therefore immediately reflected on FuelCheck's website and can be accessed by all users.

Importantly, FuelCheck in part crowdsources monitoring and enforcement by enabling consumers to report to the state government any instances whereby the prices posted at a station differ from what is posted on the

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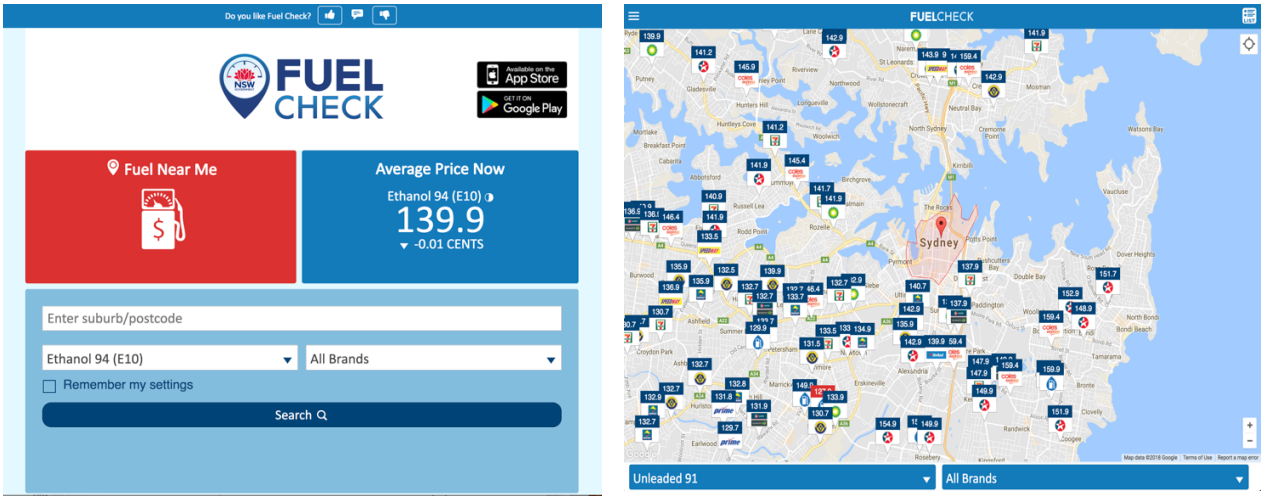
FuelCheck website. By law, they must be the same at all times. Figure 2 depicts the output from FuelCheck for Sydney's CBD, postcode 2000.

Figure 2: FuelCheck Home Page and Search Results for Sydney ULP Prices (Sydney 2000)



3. Data

In this section, we describe the data contained in the FuelWatch and FuelCheck datasets. In particular, we include the geographic coverage of both regimes, details of the variables contained in each dataset, and dimensions of the datasets.

We note from the outset that because of the differences in the structure of the FuelWatch and FuelCheck programs, the nature of their datasets differs in terms of their time frequency. FuelWatch data is only available at the daily time-frequency, which is a direct result of the 24-hour fixed price rule. In contrast, FuelCheck data is available at much higher frequencies such as the hourly, minute, or even the second-level.

3.1 Variables

Fuel types. As discussed, under both FuelWatch and FuelCheck, a station must report price changes for every type of fuel available for sale at its premises. According to FuelWatch, the full range of fuel available for sale in WA includes 7 types of fossil fuel and 1 type of bio-fossil fuel. The selection of fuel is wider in NSW, which includes 7 types of fossil fuel, 2 types of bio-fossil fuel, 1 type of bio fuel and electricity. A list of fuel types contained in FuelWatch and FuelCheck is provided in Table 1.

The availability of different fuel types on the platforms allows researchers to examine different markets for transportation, and how different market primitives such as demand elasticity affect equilibrium pricing and competitive conduct. Conceptually, different fuel types attract different types of consumers and hence are subject to different demand elasticities, which can affect competition. Examples of academic studies that exploit cross-fuel type differences in demand elasticity to examine its impact on competition include Chandra and Tappata (2011) and Isakower and Wang (2014).

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Table 1: Fuel Types Available under FuelWatch and FuelCheck

Fuel Type	FuelWatch	FuelCheck
Petrol	Unleaded 91 (ULP), Premium 95 (PULP), Premium 98 (98 RON)	Unleaded 91 (U91), Premium 95 (P95), Premium 98 (P98)
Diesel	Diesel, Branded Diesel	Diesel, Premium Diesel
Gas	Liquefied Petroleum (LPG)	Liquefied Petroleum (LPG), Compressed Natural (CNG)
Biofuel	Ethanol 85	Ethanol 94, Ethanol 105, Biodiesel 20
Other		Electric Vehicle (EV) Charge

Wholesale Prices. In studying retail competition, it is important to be able to compare retail prices to costs in order to compute a measure of profit margins. While station-specific wholesale cost data is highly proprietary and unavailable to researchers, there are publicly available wholesale cost data that are useful proxies for stations' marginal costs, and are especially useful for examining changes in margins over time. These wholesale costs are called Terminal Gate Prices, or TGP for short. These are the spot prices charged at the terminal gates where wholesale unleaded gasoline is delivered from overseas and stored. By comparing a stations' retail price to its TGP on a given date, one can obtain a measure of their margin.

FuelWatch also provides historical wholesale prices for fuel sold to downstream retailers. Wholesalers in WA are required to update their TGP prices to FuelWatch daily. In total, there are 5 wholesalers who distribute fuel from 18 petroleum terminals in WA. The earliest available data for wholesale prices dates back to December 2002. FuelCheck does not provide daily wholesale TGP's, however average wholesale prices can be obtained from the Australian Institute of Petroleum for Sydney.[2]

Variables. Tables 1 and 2 provide descriptions of the variables available in the FuelWatch and FuelCheck databases. They also provide specific examples of entries from the raw data. As can be seen in the tables, both datasets provide the exact timing a price in published, the brand and address of a given station, fuel type, and postcode. Importantly, by having the station-specific address, stations can easily be mapped using GIS programs such as ARCGIS or QGIS, which permits analyses of spatial differentiation across stations. It also allows researchers to draw concentric rings around stations, say of 2-5 kilometres, to count the number of local competitors to a station. This is common approach to defining local markets in retail gasoline studies; see, for example, Hastings (2004) or Chandra and Tappata (2011).

In addition, GIS programs can in turn map stations to local geographies that correspond to census boundaries, which in turn can allow researchers to obtain local information on demographics (e.g., income,

[2] The AIP data can be accessed at: https://www.aip.com.au/pricing/terminal-gate-prices/terminal-gate-prices-wholesale

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age, education) around a station. In practice, the finest level of publicly available demographic data from the Australian Bureau of Statistics that can be matched to a station is the "Statistical Area 1" or SA1 level. Postcode-level data, which is a more aggregate level of data, can alternatively be matched.[3]

Table 2: FuelWatch Variables

Variable Name	Explanation	Example
PUBLISH_DATE	Date when published price is offered at a station.	"1/04/2018" in dd/mm/yyyy format
TRADING NAME	Registered business name of the reporting service station	"7-Eleven Bibra Lake" "BP Augusta"
BRAND_DESCRIPTION	Service station brand	"BP" "Independent" if unbranded
PRODUCT	Fuel type	As listed in Table 1
PRODUCT_PRICE	Fuel price	"135.9" (in Australian cents)
ADDRESS	Street address of the service station	"102 Princess Rd"
LOCATION	Suburb where station is located	"BYFORD"
POSTCODE	Area postcode	"6210"
AREA_DESCRIPTION	Describes sub-areas within metropolitan Perth	"East/Hills" "North of River" "South of River"
REGION_DESCRIPTION	Describes the regional area where is the station is located.	"Metro" "Pilbara" "Kimberly"
Terminal Gate (wholesale price data only)	Name of the wholesale terminal	"BP Broome"

[3] ABS geographic area definitions, including SA1's, can be found at url below. SA1 data can be downloaded from the ABS website. http://www.abs.gov.au/websitedbs/D3310114.nsf/Home/2016%20Census%20Geography%20Basics

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Table 3: FuelCheck Variables

Variable Name	Explanation	Example
ServiceStationName	Registered business name of the reporting service station	"Caltex Brownsville" "7-Eleven Parramatta"
Address	Street address of the service station	"2355 Camden Valley Way (Corner Beech Road), Casula NSW 2170"
Suburb	Suburb where the station is located	"Parramatta"
Postcode	Area Postcode	"2289"
Brand	Service station brand	"7-Eleven" "Independent" if unbranded
FuelType	Fuel type	As listed in Table 1
PriceUpdatedDate	Timestamp of price update	"02/08/2016 06:21:05"
Price	Fuel price	"105.9" in Australian cents

Sample characteristics.

Table 4 presents summary statistics for the FuelWatch and FuelCheck samples for different fuel types: regular unleaded petrol (ULP), premium ULP, and Diesel. Focusing on regular ULP, the WA-based FuelWatch sample presented spans 16 years, from 3 Jan 2001 to 31 Dec 2017. It spans 956 stations, located in 196 postcodes, involving 25 unique petrol retailers. In total, there are 3,534,931 observations at the station-date level, which reflects the Fuelwatch 24-hour rule the restricts stations to setting just one price per day. The average daily petrol price is 125.1 cpl and the average daily margin (price less TGP) is 7.1 cpl.

The NSW-based FuelCheck sample presented covers just over one year, from 1 August 2016 to 31 Oct 2017. Compared with the FuelWatch sample, the FuelCheck sample is much shorter in terms of time span, however it is a much larger cross-section of 1,967 stations located across 494 postcodes, and is of much higher frequency. Specifically, it has 160,614 station-date observations and 233,356 unique station-minute observations. If we count all non-unique station-minute observations, as station-level prices do not change many times throughout the day (though they could potentially change in any given minute), then the FuelCheck sample has more than 1 trillion station-minute observations. The average petrol price in the sample is 122.5 cpl and the average daily margin is 6.2 cpl.

There are two other points worth noting with the FuelWatch and FuelCheck samples. First, the panels are unbalanced because of station entry and exit. Second, these samples will only continue to grow over time, and are readily updated month-to-month on the platforms' respective online historical data archives.

Table 4: Summary Statistics

Panel A: FuelWatch			
	Sample Period: 3 Jan 2001 – 31 Dec 2017		
Fuel Type	Regular (ULP)	Premium (PULP)	Diesel

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No. of unique station-date obs.	3,534,931	2,583,612	3,478,446
No. of unique stations	956	810	965
No. of unique brands	25	23	23
No. of unique dates	7,934	7,934	7,934
No. of unique postcodes	196	187	194
Average daily price (in cpl)	125.1 (20.9)	133.8 (21.6)	130.5 (23.0)
Average daily margin (in cpl)	7.1 (3.4)	6.6 (3.8)	11.6 (4.4)
Panel B: FuelCheck			
	Sample Period: 1 Aug 2016 – 31 Oct 2017		
Fuel Type	Regular (U91)	Premium (P95)	Diesel
No. of unique station-date obs,	160,614	140,083	31,794
No. of unique station-minute obs.	233,356	206,591	34,657
No. of unique stations	1,967	1,520	1,469
No. of unique brands	20	19	19
No. of unique dates	457	457	457
No. of unique postcodes	494	452	462
Average daily price (standard deviation)	122.5 (8.1)	135.5 (8.3)	124.3 (4.2)
Average daily margin (standard deviation)	12.4 (6.2)	-	13.7 (3.2)

Notes: The margin is calculated as the difference between retail and wholesale price (Terminal Gate Price). For WA, brand-specific TGP averaged over terminal locations are used to calculate average daily margins. For NSW, U91 and Diesel TGPs are sourced from the Australian Institute of Petroleum. Standard deviations are in parentheses.

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4. Petrol Price Movements

This section provides some visual descriptions of the FuelWatch and FuelCheck petrol price data. We consider different time frequencies to highlight the different types of patterns observed in the data: monthly, daily, hourly, and real-time. For each type of time series plot, we also discuss areas of research that are typically examined with the data.

Monthly price movements. Figures 3 and 4 plot average monthly retail regular gasoline[4] prices across all stations in Perth and Sydney computed using the raw FuelWatch and FuelCheck data. We also plot the monthly average of the wholesale TGP in Perth, which highlights movements in international crude oil prices; all other TGP's at all terminal gates across Australia exhibit similar patterns.

From the figure, it is clear that retail prices move closely with TGP's over time. As the summary statistics above showed, the average gap between the retail prices and TGP in the figure is 7 cpl and 6 cpl for Perth and Sydney, respectively. This provides a sense of the magnitude of retail margins earned in the retail petrol market. The figure also highlights collapses in retail and wholesale petrol prices in 2008-09 and 2015. The former collapse is often attributed to a negative oil demand shock in China (Hamilton 2009), while the latter collapse is related to a global supply glut in the Middle East and shale oil drilling in the United States.

The literature that focuses on monthly petrol price movements is that which examines that rate at which wholesale cost shocks are passed through to retail prices. A substantial body of research dating back to Bacon (1991) and Borenstein et al. (1997) has examined the "rockets and feathers" phenomenon with retail price passthrough of cost shocks in gasoline, where by prices asymmetrically rise quickly in response to increases in costs, while they fall slowly in response to decreases in costs. An on-going area of research continues to examine how market power (e.g., Marvel 2008; Byrne 2018) and consumer search behaviour (e.g. Lewis and Marvel 2013) generates such asymmetric patterns in cost passthrough. Separate from cost passthrough, researchers could also potentially use monthly petrol price data and wholesale costs to potentially investigate the impact of fuel cost shocks on household savings and investment behaviour, if these latter types of data are matched to petrol price data.

[4] ULP in Perth and U91 in Sydney

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Figure 3: Monthly Average Retail and Wholesale ULP Prices in Perth (2001 – 2017)

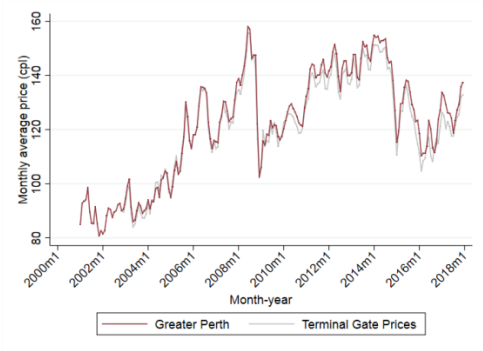
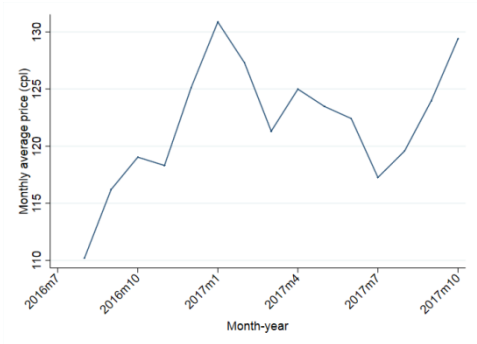


Figure 4: Monthly Average Retail U91 Prices in Sydney (2016 – 2017)



Daily price movements. Figures 5a and 5b plot daily average retail prices across 46 and 502 stations in Perth (from February 2001 to December 2017) and Sydney (from August 2016 to September 2017). What is striking in both figures is that petrol prices exhibit cycles where they periodically jump and then exhibit periods of undercutting between the jumps. Indeed, petrol price cycles exist in all major cities across Australia (Byrne 2014), as well as in international cities in the U.S., Canada, and Europe (Eckert 2013). The figures also show that the cycles have different lengths in the two cities, and that cycles do not exist for diesel.

In contrast, Figure 6 plots daily average retail prices across stations in regional WA and regional NSW for the same August 2016-September 2017 sample period. In contrast to Perth and Sydney, we do not find cycles at all, and instead see highly rigid prices. Byrne (2018) finds similarly in regional Canadian markets.

These time series naturally raise the following questions: why and when do we see petrol price cycles? While research into petrol price cycles dates back to Castanias and Johnson (1993), it remains an open question as to why cycles exist, and whether they are good or bad for consumers. Using daily price data, researchers have recently examined the impact of price cycles on cost passthrough (Lewis and Noel, 2011), the role of price leadership in coordinating cycles (Lewis, 2012; Byrne et al., 2015), and whether they play a collusive function (Clark and Houde, 2013) or arise from competition among asymmetric retailers (Eckert 2003; Noel 2007). This remains an open area of research where, presumably, hourly-level or even higher frequency price data like that available from FuelCheck (discussed next) will enable new insights into the causes and consequences of petrol price cycles. On the demand-side of the market, there is little evidence on consumer

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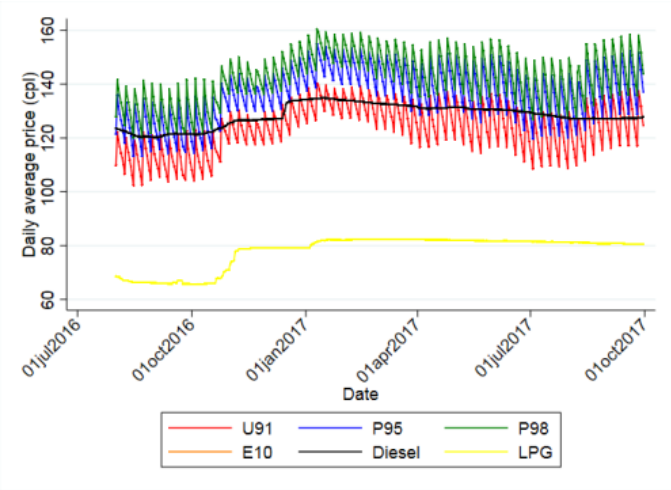
demand responses to petrol price cycles with recent exceptions being Byrne et al (2015), Byrne and de Roos (2017), and Noel (2018). This is yet another area for potential future research.

Figure 5: Daily Averages of Fuel Prices in Perth and Sydney

a) Perth



b) Sydney

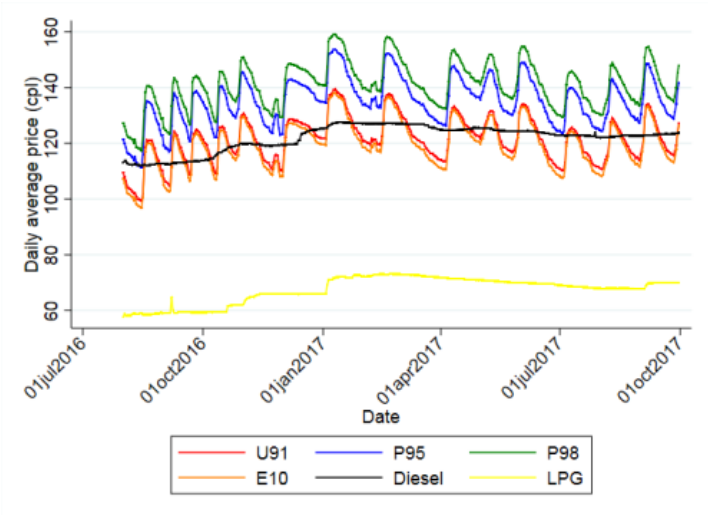


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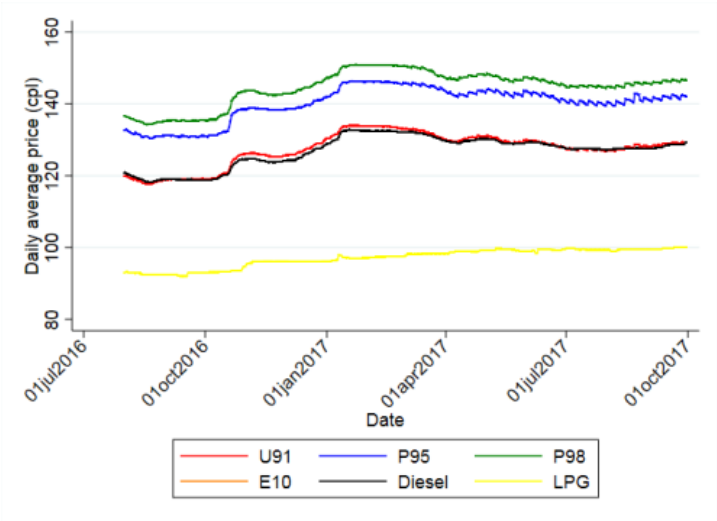
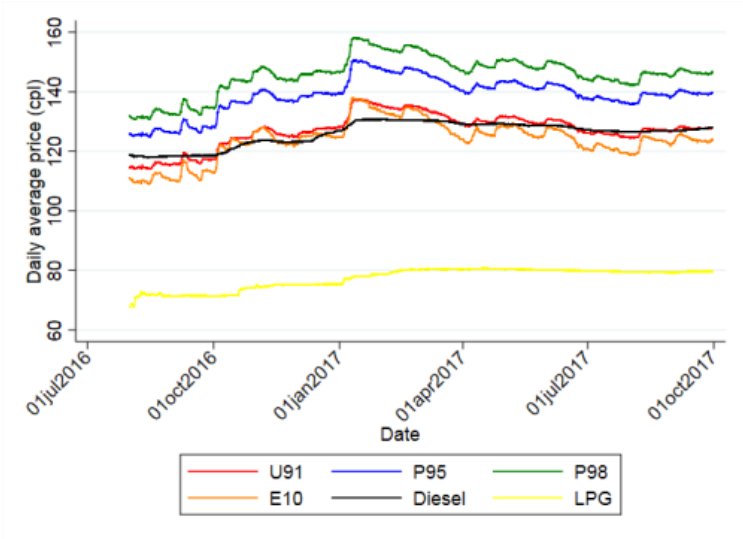
Figure 6: Daily Averages of Fuel Prices in Regional Western Australia and New South Wales

a) Regional Western Australia



b) Regional New South Wales



Real-time price movements. Figures 7 and 8 present our final two sets of figures, which highlight the richness of even higher frequency and finer aggregation price data from FuelCheck. More specifically, Figure 7 presents hourly average prices for BP, Caltex, Coles, and Woolworths stations in Sydney for the dates 2 January 2017 to 7 January 2017, or for 6 days in total. Here, we can see the timing by hour-of-the-day over this period with which brands engage in price jumps in Sydney over this period as they coordinate on a new price level of approximately 143 cpl at the top of the cycle, rising from 123 cpl at the bottom of the previous cycle.

Figure 8 digs even deeper into the data, and presents real-time station-specific price changes by minute of the day for the 4th of January 2017 to 6th of January 2017 period for the postcode of Taren Point (2229) in Sydney. Here, we can see the exact timing, ordering, and magnitude of price changes among these stations. We see in Figure 7 before that prices start to relent within a few hours to the peak of the cycle across the

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market. By plotting price updates in Taren Point in Figure 8, we see that Coles Express Taren Point leads the price jump at precisely 7.40 am on the 5th of January 2017, and is followed two and half hours later by 7-Eleven at 10.12am.

There are few studies to date that exploit petrol price data of this high a frequency and at this level of disaggregation to examine oligopoly pricing. Wang (2009) uses data from FuelWatch to examine the impacts of FuelWatch's introduction on price leadership and mixed strategy pricing equilibria in Perth. Byrne and de Roos (2018) use the FuelWatch data to investigate and equilibrium transition between 2009 and 2013 led by BP, who uses price leadership, price experiments, and gradualism to create pricing focal points and enhance margins.

As these two studies show, such rich price data are lucrative for shedding new light on the oligopoly problem in practice. We expect there are many more such intensely-micro empirical studies that could be developed going forward, especially with the FuelCheck data, which is of even higher frequency and covers even more local heterogeneous markets across NSW (as compared to FuelCheck in less-populated WA). Studying changes in oligopoly pricing and equilibrium in the market around major technological or policy shocks is of particular interest since these are often revealing of the underlying sources of strategic interaction and the informational constraints facing firms in setting prices. For instance, mergers are major market structure shocks that have proven useful for such investigations in earlier research with less rich data (Hastings 2004).

Figure 7: Hourly Petrol Prices in Sydney by Brand

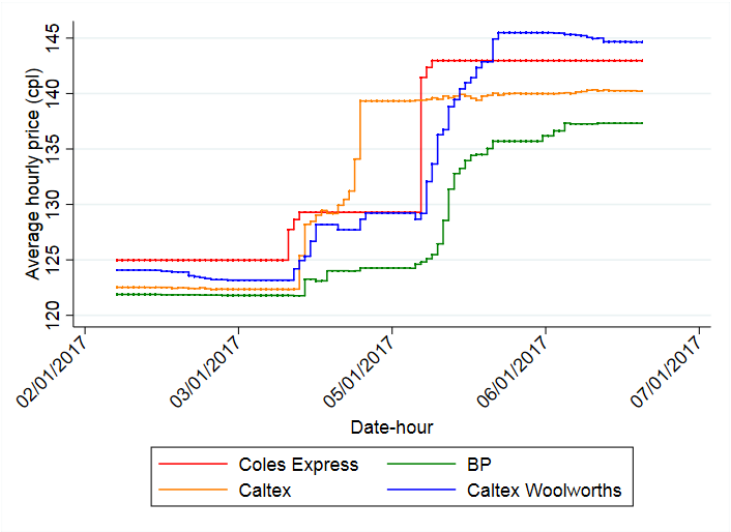

Figure 8: Hourly Station-Level Petrol Prices in Taren Point, 2229

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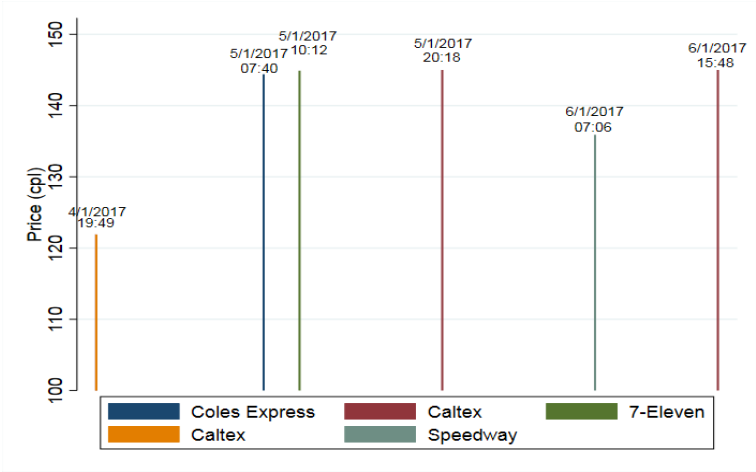

Price dispersion. Throughout this overview of the FuelWatch and FuelCheck data, we have focused on price levels in illustrating the types of patterns exhibited in the data. There is, however, and equally important area of research that could be investigated with these rich data: price dispersion across stations, and over time. For instance, price dispersion in petrol markets is typically measured by the standard deviation or range of prices. Indeed, Figures 2 and 3 above highlight dispersion in prices across stations in Perth and Sydney. Petrol markets are great contexts for examining such dispersion since the product being sold itself – fuel – is homogeneous. This implies petrol markets are align well with standard models of consumer search and price dispersion, which assume product homogeneity (Varian 1980) and focus on consumer search frictions as an underlying source of price dispersion across oligopolists.

Again, the richness of the FuelWatch and FuelCheck data over time and space permit new analyses of the underlying sources of price dispersion in retail markets, including spatial frictions, traffic congestion, shopper docket discounts from supermarket purchases, and so on.[5] In the context of price transparency policies, such studies are particularly important since the primary goal of these policies are to reduce consumer search costs to promote competition. We refer the interested reader to Eckert (2013) article for an overview of the empirical literature on search and price dispersion in petrol markets for further background on such studies.

5. Conclusion

FuelWatch and FuelCheck are among the world's best sources for petrol price data, permitting research that is of the highest academic and policy relevance. Academically, petrol markets allow researchers to investigate fundamental issues in industrial organisation including price leadership, mergers, collusion, price dispersion, consumer search, among various other pertinent issues in industrial organisation. Going forward, The richness of the datasets, particularly FuelCheck, should permit new tests of theories and reveal new empirical patterns to inform theories of firm and consumer behaviour.

On the policy front, price transparency policies have emerged in petrol markets around the world. More generally, they have also emerged in supermarkets, healthcare markets, superannuation, and other important

[5] Pennerstorfer et al. (2017) is an example of a recent article that exploits rich petrol price data from a price transparency platform in Germany to develop a novel micro-examination of search frictions and price dispersion based on commuter flows.

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industries where consumers face search frictions, and where worries of market power and inequities persist. FuelWatch and FuelCheck represent rich new data sources for investigating how consumers and firms react to the presence of price transparency platforms, which can lead to both intended and potentially unintended consequences.

Finally, we close by reiterating that these microdata are *free* and can be easily downloaded from the platforms' websites. This fact, combined with the many fundamental issues in industrial organisation can be examined in the context of petrol markets, implies that the FuelWatch and FuelCheck datasets are terrific source of information for economic students to develop Honours or Master's essays or PhD dissertations. Typically studies in empirical industrial organisation rely on expensive, proprietary data, which represents a major barrier for entry for economics students. By making realtime petrol price data available, the WA and NSW governments have removed these restrictions and created a multitude of research opportunities for students going forward.

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